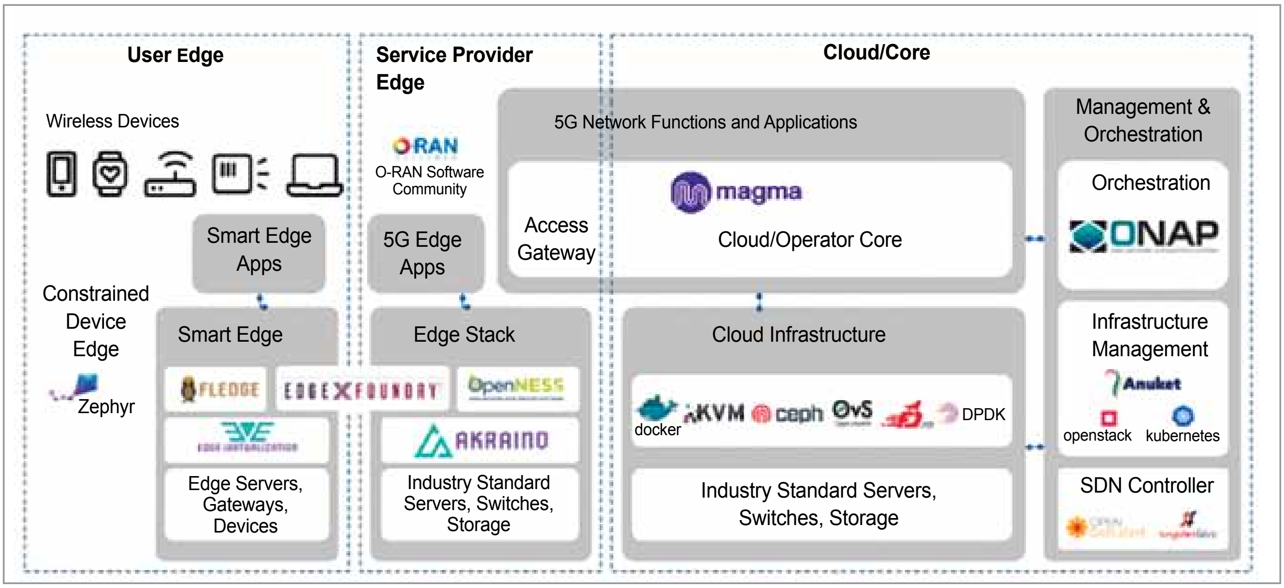

Figure 1: LF open source component projects for 5G

While there are projects like Anuket, which is in LF networking, projects which are LFH can solve exactly this problem. They look at blueprints and how to comply with a spec, do open verification, submit the test results and do open interrupt, testing, training, and certification. So that's kind of how we are seeing most of the evolution in open source this year.

Collaboration

LF networking has a unique perspective with the broadest set of projects that make up the diverse and evolving open source networking stack. It provides platforms and building blocks across the networking industry that enable rapid interoperability, deployment, and adoption, and is the nexus for 5G innovation and integration. It has now tapped into industry efforts to structure a new initiative to develop 5G Super Blueprints for the ecosystem. Major integrations between the building blocks are now underway between ONAP and ORAN, Akraino and Magma, Anuket and Kubernetes, and more.

The 5G super blueprint is what is going to integrate a set of projects all the way from access to core. 'Super' means that we are integrating multiple projects, umbrellas (such as LF Edge, Magma, CNCF, O-RAN Alliance, and LF Energy) with an end-to-end structure for the underlying facilities and application layers throughout edge, gain access to, and core. This end-to-end combination allows leading market usage cases such as repaired wireless, mobile broadband, personal 5G, multi-access, IoT, voice services, and network slicing. In short, 5G Super Blueprints are a vehicle to team up and produce end-to-end 5G options.

With availability of customised solutions, there is a tremendous opportunity for the market. This is how the verticals are taking advantage of edge computing. Now healthcare and smart cities are towards the lower end of the spectrum. But it will need a little bit of regulations and process controls to update before adoption happens.

LF edge projects

There are two types of edges—user edge and service provider edge. A user edge is dedicated and operated by the user. A service provider edge is shared as a service, but then not all usages are the same. You can either have a very constrained device edge, which could be microcontroller based embedded compute, or you could have a gateway type edge. You know in the user control or an on-prem data centre edge, which is secure in a factory building.

And what separates the two is the last mile networks. So, under the base station or in a smart central office, which is the access edge is not hard to know. Edge compute is for applications that need 20 milliseconds or less latency.

The edge project stack is a very important attribute. For example, if a sensor wakes up and dumps data to an Amazon cloud every week, it's an IoT application. It's not an edge application, and vice versa. Hence, the LF edge as an umbrella is unifying the frameworks.

Akraino R4 blueprints

Akraino is a blueprint concept. The open community comes together to define a use case. And then, through open governance, tests out the use case and interoperability all the way from hardware to the application,